One problem was the timescale. SEI extended five or so presidential terms of office into the future (taking the average presidency as one and a half terms). That makes it easy for a president to attempt to commit his successors, but leaves in considerable doubt how reliable such a commitment might be. SEI contrasted dramatically with the *Apollo* program—which, it might have been conjectured at the time it began, could have triumphed when President Kennedy or his immediate political heir was still in office.

Second, there was concern about whether NASA, which had recently experienced great difficulty in safely lifting a few astronauts 200 miles above the Earth, could send astronauts on an arcing year-long trajectory to a destination 100 million miles away and bring them back alive.

Third, the program was conceived exclusively in nationalist terms. Cooperation with other nations was not fundamental to either design or execution. Vice President Dan Quayle, who had nominal responsibility for space, justified the space station as a demonstration that the United States was "the world's only superpower." But since the Soviet Union had an operational space station that was a decade ahead of the United States, Mr. Quayle's argument proved difficult to follow.

Finally, there was the question of where, in terms of practical politics, the money was supposed to come from. The costs of getting the first humans to Mars had been variously estimated, ranging as high as $500 billion.

Of course, it's impossible to predict costs before you have a mission design. And the mission design depends on such matters as the size of the crew; the extent to which you take mitigating steps against solar and cosmic radiation hazards, or zero gravity; and what other risks you are willing to accept with the lives of the men and women on board. If every crew member has one essential specialty, what happens if one of them falls ill? The larger the crew, the more reliable the backups. You would almost certainly not send a full-time oral surgeon, but what happens if you need root canal